

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: CST203**Course Name: Logic System Design**

Max. Marks: 100

Duration: 3 Hours

PART A**Answer all questions. Each question carries 3 marks**

Marks

- 1 Do the following base conversions (3)
 - a) $(96DE)_{16}$ to octal
 - b) $(1011011000)_2$ to octal
- 2 Subtract -12 from 23 using 2's complement representation and 1's complement representation (3)
- 3 State and prove extended De Morgan's theorem (3)
- 4 Using Huntington's postulates prove that (3)
 - a) $x + x = x$
 - b) $x + 1 = 1$
- 5 Distinguish between decoder and demultiplexer (3)
- 6 Design a half adder circuit from its truth table (3)
- 7 Distinguish between T flip-flop and D flip-flop (3)
- 8 Explain race around problem. How can it be eliminated? (3)
- 9 Write the algorithm for addition of two binary numbers in 2's complement form (3)
- 10 What is programmable logic array? Where is it useful? (3)

PART B**Answer any one full question from each module. Each question carries 14 marks****Module 1**

- 11 a) Convert i) $(214)_{10}$ to binary, octal, BCD and hexadecimal (4)
 - ii) (128) to binary, octal, BCD and hexadecimal (4)
- b) Represent -219 and -114 in (6)
 - i) sign magnitude form ii) 1's complement form
 - iii) 2's complement form
- 12 a) Add 127 and 765 assuming the numbers are i) octal ii) BCD iii) hexadecimal (6)
- b) Subtract 157 from 615 assuming the numbers are i) octal ii) BCD (6)
 - iii) hexadecimal iv) 2's complement form (2)

Module 2

- 13 a) Define Boolean algebra. Give an example (8)
b) Show that any digital circuit can be implemented using universal gates (6)
- 14 a) Simplify the Boolean function $F(a,b,c,d) = \sum (0,1,2,5,7,8,9,10,11,13,15)$ using K map (7)
b) Verify your answer using tabulation method. (7)

Module 3

- 15 a) Explain parallel adder/subtractor circuit with a logic diagram (8)
b) Design a carry look ahead adder circuit for four bit binary addition (6)
- 16 a) Design a code converter circuit for converting binary number to BCD number (8)
b) Design a 4x2 encoder circuit (6)

Module 4

- 17 a) Explain 3 bit binary asynchronous counter with a logic diagram and timing sequence (8)
b) Explain asynchronous BCD counter (6)
- 18 a) Explain i) SR flip-flop ii) JK flip-flop iii) master-slave flip-flop with excitation table and characteristic equation (12)
b) Explain edge triggered flip-flop (2)

Module 5

- 19 a) Explain a ring counter with a logic diagram, timing sequence and state diagram (10)
b) Explain with a logic diagram a serial in parallel out shift register (4)
- 20 a) Illustrate the algorithm for addition and subtraction of two BCD numbers with an example (8)
b) Explain with an example how simple functions can be implemented using PLA (6)
